

Measured-only time-calibrated *Desmognathus* phylogeny with audited trait uncertainty

Genome panel: 721 reviewed, image-quality-matched nuclei (relative IOD, not pg). Size panel: 50 manually vetted largest cells and their corresponding nuclei per species. No phylogenetic fills.

Descriptive species-level Spearman ρ : IOD–nucleus 0.45; IOD–cell 0.55; nucleus–cell 0.66. These correlations are not phylogenetically corrected.

Time-calibrated phylogeny

Relative nuclear-IOD

Nucleus area

Cell area

Bootstrap estimator distribution Current point estimate Across-species median

